

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2022 - 2023)

JEE(Main)

UNIT TEST

18-09-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-2

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

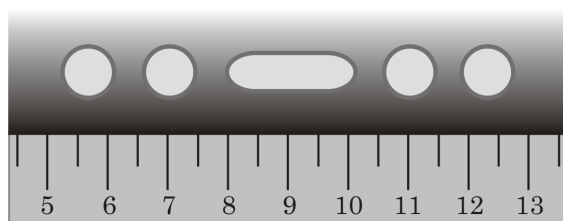
1. A 750-nm light beam hits the flat surface of a certain liquid, and the beam is split into a reflected ray and a refracted ray. If the reflected ray is completely polarized and angle of refraction is 37° , what is the wavelength of the refracted ray ?
 (A) 1000 nm (B) 562.5 nm
 (C) 500 nm (D) 1125 nm
2. The path difference between two interfering waves at a point on the screen is $\lambda/6$. The ratio of intensity at this point and that at the central bright fringe will be : (Assume that intensity due to each slit is same)
 (A) 0.853 (B) 8.53
 (C) 0.75 (D) 7.5
3. If we study the vibration of a pipe open at both ends, then the following statements is not true :
 (A) Only odd harmonics of the fundamental frequency will be generated.
 (B) All harmonics of the fundamental frequency will be generated.
 (C) Open end will be pressure node
 (D) Open end will be displacement antinode

4. A standing wave having 3 nodes and 2 antinodes is formed between two atoms having a distance of $1.21 \text{ \AA}$ between them. The wavelength of the standing wave is -
 (A) $1.21 \text{ \AA}$
 (B) $2.42 \text{ \AA}$
 (C) $3.63 \text{ \AA}$
 (D) $6.05 \text{ \AA}$
5. A closed organ pipe and an open organ pipe are tuned to the same fundamental frequency. What is the ratio of their lengths ?
 (A) 1 : 2 (B) 2 : 1
 (C) 2 : 3 (D) 4 : 3
6. A traveller is running with velocity v_0 towards a stationary train. If the train sounds a horn of frequency n then the apparent frequency heard by the traveller will be (v is speed of sound) :
 (A) $n' = n \left(\frac{v + v_0}{v} \right)$
 (B) $n' = n \left(\frac{v - v_0}{v + v_s} \right)$
 (C) $n' = n \left(\frac{v - v_0}{v} \right)$
 (D) None of these
7. The equation of a progressive wave is $y = 0.02 \sin 2\pi \left[\frac{t}{0.01} - \frac{x}{0.3} \right]$, where x and y are in meters and t is in second. The velocity (in m/s) of propagation of the wave is :
 (A) 400 (B) 40 (C) 300 (D) 30

8. In a sinusoidal wave, the time required for a particular point to move from maximum displacement to zero displacement is 0.17 sec. The frequency of the wave is :-

(A) 1.47 Hz
(B) 0.36 Hz
(C) 0.73 Hz
(D) 2.94 Hz

9. Laser light with a wavelength of 500 nm is directed through one slit or two slits and allowed to fall on a screen 2.60 m beyond. Figure shows the pattern on the screen, with a centimeter ruler below it. Did the light pass through one slit or two slits ? If one slit, find its width. If two slits, find the distance between their centers.



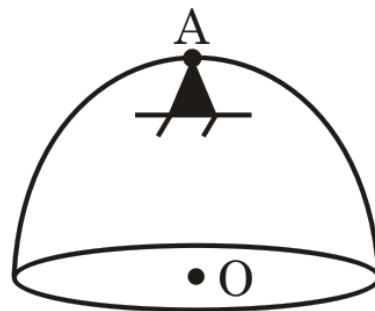
- (A) One slit of width approx 0.1 mm.
(B) One slit of width approx 0.05 mm.
(C) Two slits at an approx. distance of 0.08 mm
(D) Two slits at an approx. distance of 0.04 mm
10. The antenna of a 6 kW radio station radiates spherical wavefronts. At a distance of 100 m from the antenna, what is the peak magnetic field strength ?

(A) 20 nT (B) $10\sqrt{2}$ nT
(C) 10 nT (D) $\frac{5}{\sqrt{2}}$ nT

11. A particle executes SHM and is located at $x = a$, $x = b$ and $x = c$, at $t = t_0$, $t = 2t_0$ and $t = 3t_0$ respectively. The angular frequency of oscillation is

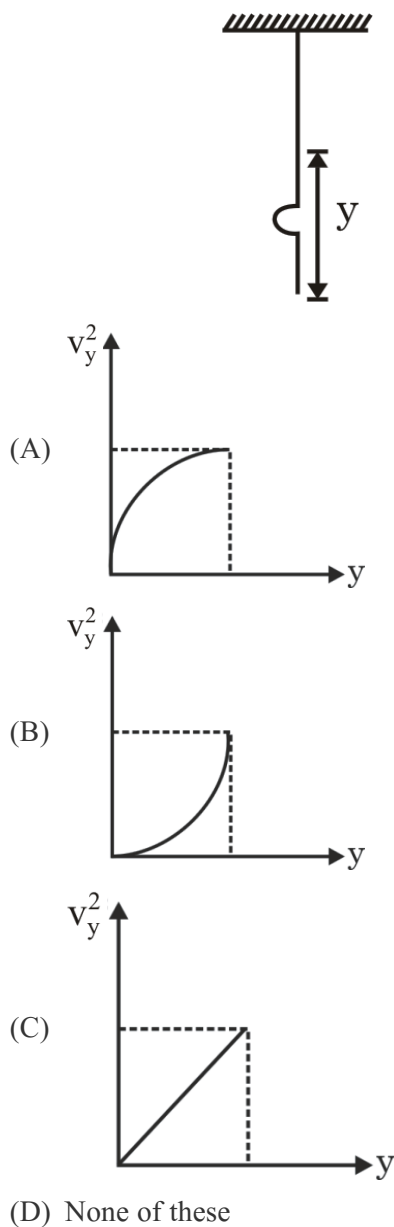
(A) $\frac{1}{t_0} \sin^{-1} \left(\frac{a+c}{2b} \right)$
(B) $\frac{1}{t_0} \sin^{-1} \left(\frac{a+c}{b} \right)$
(C) $\frac{1}{t_0} \cos^{-1} \left(\frac{a+c}{b} \right)$
(D) $\frac{1}{t_0} \cos^{-1} \left(\frac{a+c}{2b} \right)$

12. A uniform hemispherical shell of mass M and radius R is made to perform small angular oscillation about point 'A' such that line AO always remains in vertical plane (where O is curvature centre of spherical shell). Frequency of small angular oscillations will be :

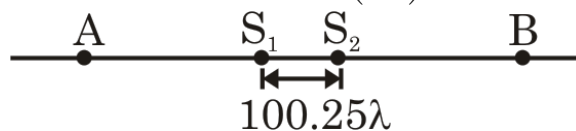


(A) $\frac{1}{2\pi} \sqrt{\frac{6g}{R}}$
(B) $\frac{1}{2\pi} \sqrt{\frac{3g}{R}}$
(C) $\frac{1}{4\pi} \sqrt{\frac{3g}{R}}$
(D) $\frac{1}{4\pi} \sqrt{\frac{g}{R}}$

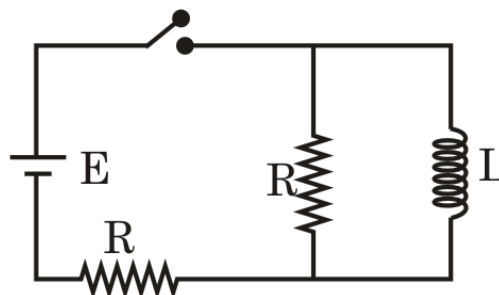
13. A non-uniform rope of length ℓ hangs from a ceiling. Mass per unit length of rope (μ) changes as $\mu = \mu_0 e^y$, where y is the distance along the string from its lowest point. Then graph between square of velocity of wave (v_y^2) and y will be best represented as :



14. S_1 and S_2 are two coherent sources of radiations separated by distance 100.25λ , where λ is the wave length of radiation. S_1 leads S_2 in phase by $\frac{\pi}{2}$. A and B are two points on the line joining S_1 and S_2 as shown in figure. The ratio of amplitudes of component waves from source S_1 and S_2 at A and B are in ratio 1 : 2. The ratio of intensity at A to that of B ($\frac{I_A}{I_B}$) is :

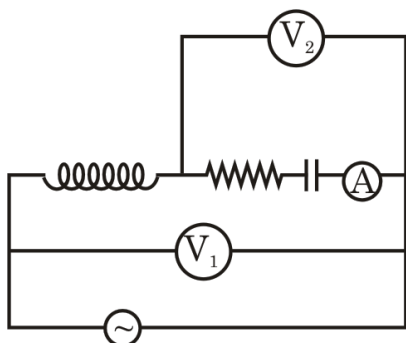


- (A) ∞
 (B) $\frac{1}{9}$
 (C) 0
 (D) 8.1
15. In the given circuit key is closed at $t = 0$. At what time the P.D. across inductor is one fourth of emf of the cell.

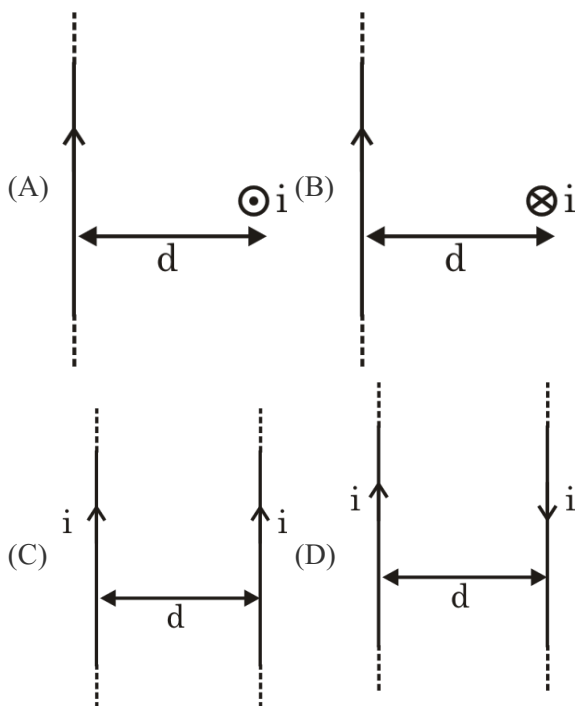


- (A) $\frac{L}{R} \ln 2$
 (B) $\frac{L}{R} \ln 4$
 (C) 0
 (D) $\frac{2L}{R}$

16. In the ac circuit shown, $X_L = 7\Omega$, $R = 4\Omega$ and $X_C = 4\Omega$. The reading of the ideal voltmeter V_2 is $8\sqrt{2}$ V. The reading of the ideal ammeter is.



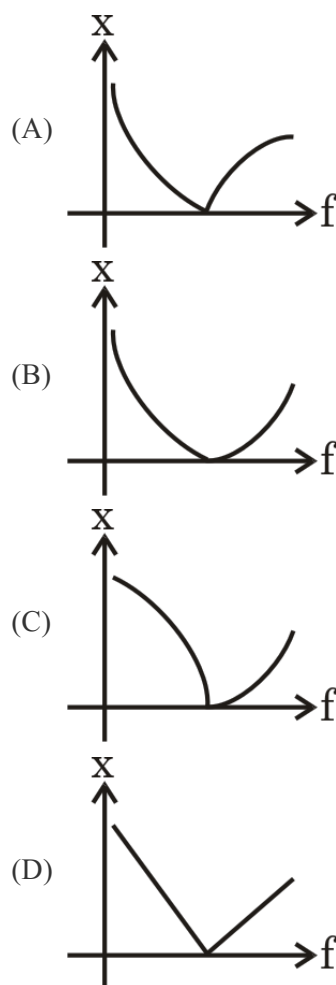
- (A) 3A
(B) 2A
(C) 4A
(D) 5A
17. The arrangements of wires in which there exists atleast one point at finite distance from wires, at which magnetic field is zero.



18. If the earth's field induction at a place is 0.36 gauss and the angle of dip is 60° then the horizontal and vertical component of the field is :

- (A) 0.36 gauss, $0.36\sqrt{3}$ gauss
(B) 0.18 gauss, $0.18\sqrt{3}$ gauss
(C) 0.09 gauss, $0.09\sqrt{3}$ gauss
(D) None of these

19. In a series L, C circuit, which of the following represents variation of magnitude of reactance (X) with frequency (f)



20. Three copper wires of length and cross-sectional area (L, A) , $(2L, A/2)$ and $(L/2, 2A)$. Resistance is minimum in
- (A) wire of cross-sectional area A
- (B) wire of area $A/2$
- (C) wire of cross-sectional area $2A$
- (D) same in all three cases.

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is an Integer Value Type questions.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

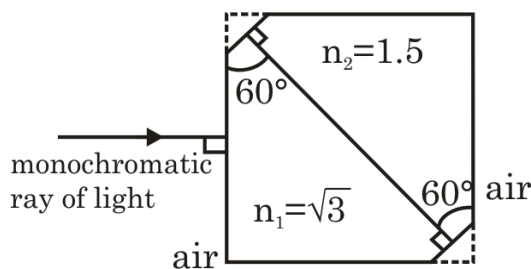
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

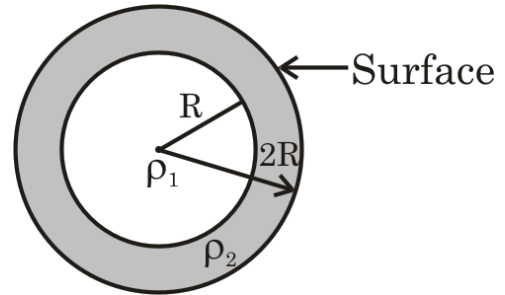
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

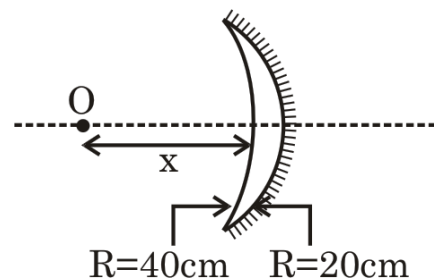
1. Refractive indices of 2 different media with separating boundary at the diagonal of rectangular glass slab are shown. Then angle of deviation (in degree) of the ray in going from n_1 to n_2 as shown in the figure is :



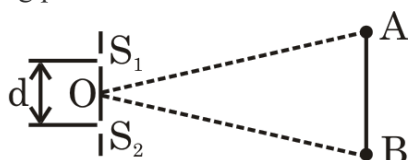
2. A planet is made of two materials of density ' ρ_1 ' and ' ρ_2 ' as shown in figure. Acceleration due to gravity at surface of planet is same as that at depth ' R '. The ratio $\frac{\rho_1}{\rho_2}$ is equal to $\frac{x}{3}$. Find the value of ' x '.



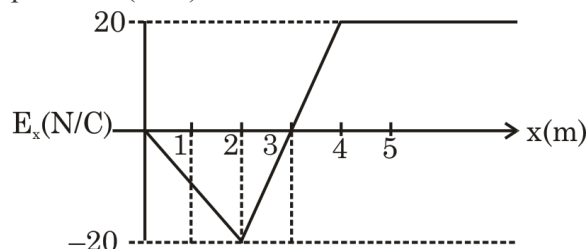
3. The potential energy of a particle of mass m free to move along x axis has the form $U(x) = \frac{a}{x^2} - \frac{b}{x}$, where a and b are positive constant and x is x -coordinate of the particle. The time period of SHM (along x -axis) of small amplitude around mean position is $\frac{P\pi a}{b^2} \sqrt{2ma}$. Find the value of P .
4. Radii of curvature of a concavo-convex lens (refractive index = 1.5) are 40 cm (concave side) and 20 cm (convex side) as shown. The convex side is silvered. The distance x on the principal axis where object is placed so that its image is created on the object itself, is equal to (in cm) :



5. The frequency of a fork A is 3% more than the frequency of a standard fork whereas the frequency of fork B is 3% less. The forks A and B produce 6 beats per second. The frequency of standard fork (in Hz) will be :-
6. If the ratio of the intensity of two coherent sources is 4 then the visibility $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$ of the fringes is n . Find the value of $10n$.
7. Figure shows two coherent sources S_1 and S_2 vibrating in same phase, AB is an wire lying symmetrically at a far distance from the sources S_1 and S_2 . Let $\frac{\lambda}{d} = 10^{-3}$. $\angle BOA = 0.12^\circ$. How many bright spots will be seen on the wire, including points A and B.



8. A graph of the x component of the electric field as a function of x in a region of space is shown. The Y and Z components of the electric field are zero in this region. If the electric potential is 10V at the origin, then potential (in V) at $x = 2.0$ m is :



9. Consider a uniformly charged solid non conducting sphere of radius R and charge Q . Consider three spherical regions centered at the centre of the uniformly charged solid non conducting sphere, as given below :
- Region (1) $R_1 \geq r \geq 0$
- Region (2) $R_2 \geq r \geq R_1$
- Region (3) $\infty \geq r \geq R_2$
- (Both R_1 & $R_2 > R$)
- If electrostatic potential energy stored in each of the three regions is equal then $\frac{R_2}{R_1}$ is :
10. A sound absorber attenuates the sound level by 20 dB. The intensity decreases by a factor of :

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

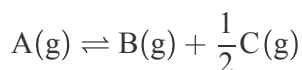
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Select the incorrect statement about three-dimensional HCP crystal made of identical atoms :
[Given : r = radius of atom]

- (A) The number of atoms in HCP unit cell is six
(B) The volume of HCP unit cell is $24\sqrt{2}r^3$
(C) The empty space in HCP unit cell is 26%
(D) The base area of HCP unit cell is $1.5\sqrt{3}r^2$

2. For a reaction at equilibrium



the relation between dissociation constant (K), degree of dissociation (α) and equilibrium pressure (p) is given by :

(A) $K = \frac{\alpha^{\frac{1}{2}} p^{\frac{3}{2}}}{(1 + \frac{3}{2}\alpha)^{\frac{1}{2}} (1 - \alpha)}$

(B) $K = \frac{\alpha^{\frac{3}{2}} p^{\frac{1}{2}}}{(2 + \alpha)^{\frac{1}{2}} (1 - \alpha)}$

(C) $K = \frac{(\alpha p)^{\frac{3}{2}}}{(1 + \frac{3}{2}\alpha)^{\frac{1}{2}} (1 - \alpha)}$

(D) $K = \frac{(\alpha p)^{\frac{3}{2}}}{(1 + \alpha) (1 - \alpha)^{\frac{1}{2}}}$

3. Enthalpy of neutralization of H_3PO_3 acid is -106.68 kJ/mol using NaOH. Enthalpy of neutralization of HCl by NaOH is -55.84 kJ/mol. Calculate $\Delta H_{\text{ionization}}$ of H_3PO_3 into its ions:

- (A) 50.84 kJ/mol (B) 5 kJ/mol
(C) 2.5 kJ/mol (D) None of these

4. The enthalpy of combustion of propane (C_3H_8) gas in terms of given data is :

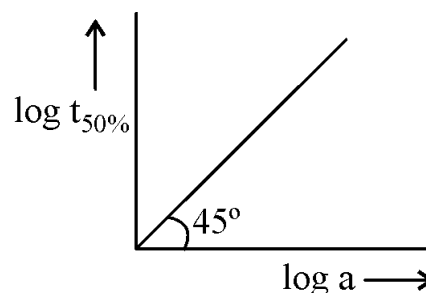
Bond energy (kJ/mol)

ϵ_{C-H}	$\epsilon_{O=O}$	$\epsilon_{C=O}$	ϵ_{O-H}	ϵ_{C-C}
$+x_1$	$+x_2$	$+x_3$	$+x_4$	$+x_5$

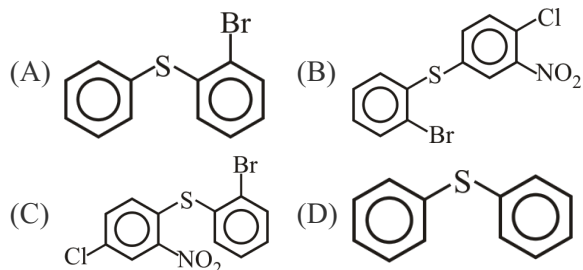
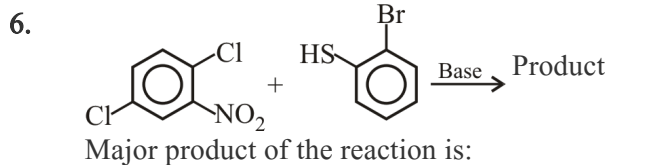
Resonance energy of CO_2 is $-z$ kJ/mol and $\Delta H_{\text{vaporization}} [H_2O(l)]$ is y kJ/mol.

- (A) $8x_1 + 2x_5 + 5x_2 - 6x_3 - 8x_4 - 4y - 3z$
(B) $6x_1 + x_5 + 5x_2 - 3x_3 - 4x_4 - 4y - 3z$
(C) $8x_1 + 2x_5 + 5x_2 - 6x_3 - 8x_4 - y - z$
(D) $8x_1 + x_5 + 5x_2 - 6x_3 - 8x_4 - 4y + 3z$

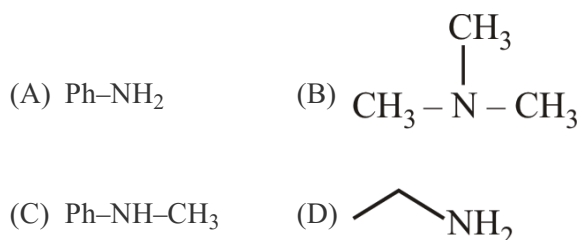
5. What will be the order of reaction $[A \rightarrow P]$ and rate constant for this chemical change having $[\log t_{50\%}]$ vs $[\log \text{initial concentration of A (a)}]$ curves as :



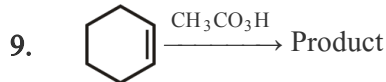
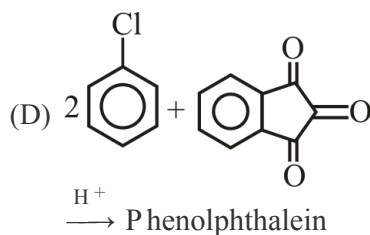
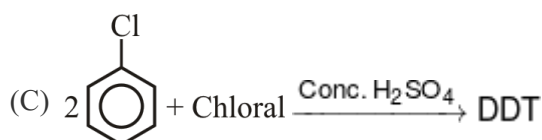
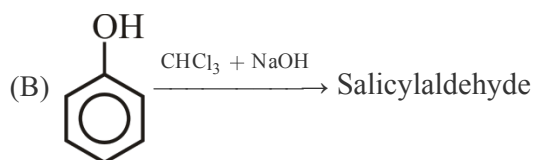
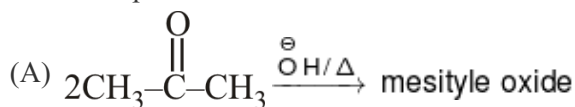
- (A) 0, 1/2 (B) 1, 1
(C) 2, 2 (D) 3, 1



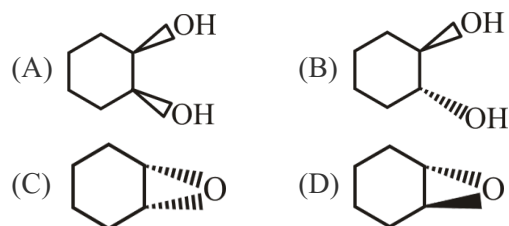
7. Which of the following Amine can be synthesised in good yield by Gabriel phthalimide synthesis?



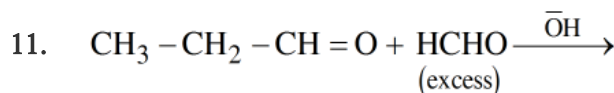
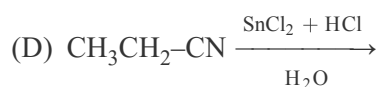
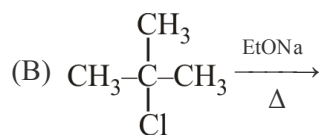
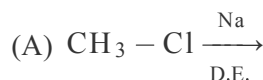
8. Which of the following is incorrectly matched with their product?



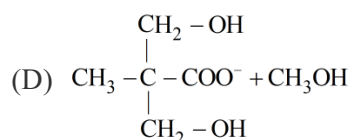
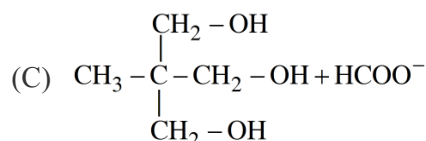
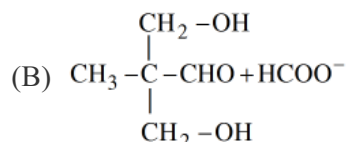
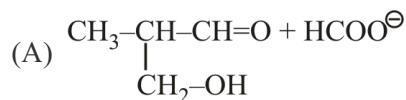
Product formed in the reaction is -



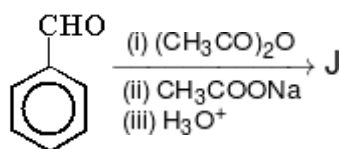
10. Which of the following is supposed to produce alkene



Which of the following product mixture is formed as a final major product

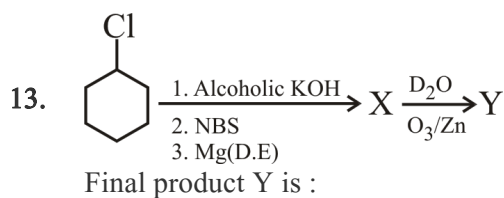
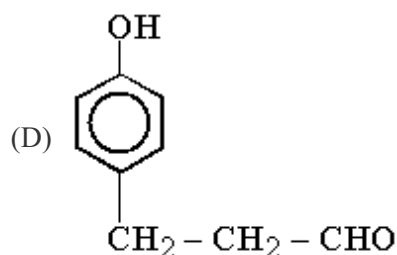
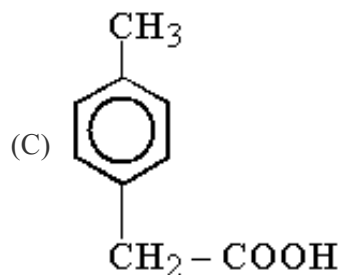
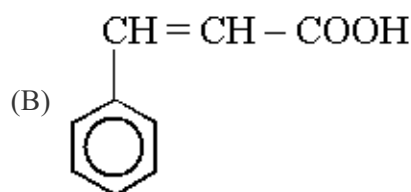
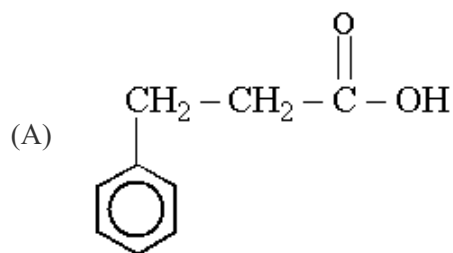


12. In the following reaction, the compound J is :



J gives effervescence on treatment with NaHCO_3 and a positive Baeyer's test.

The compound J is :



- (A) $\text{O}=\text{CH}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}=\text{O}$
(B) $\text{O}=\text{CH}-\underset{\text{D}}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}=\text{O}$
(C) $\text{O}=\text{CH}-\text{CH}_2-\underset{\text{D}}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{CH}=\text{O}$
(D) $\text{O}=\text{CH}-\underset{\text{D}}{\text{CH}}-\text{CH}_2-\text{CH}_2-\underset{\text{D}}{\text{CH}}-\text{CH}=\text{O}$

14. Choose **CORRECT** statement(s) regarding borax :

- (i) Its aqueous solution is acidic in nature
(ii) It is prepared by using colemanite mineral
(iii) Its aqueous solution acts as buffer solution
(iv) It produces brick red transparent bead on heating with CaCO_3

- (A) (i), (ii)(iii) only
(B) (ii), (iii) only
(C) (ii), (iii)(iv) only
(D) All of these

15. Which of the following statement is correct for H_2O and D_2O ?

- (A) D_2O has higher number of valence shell electron than H_2O
(B) D_2O has lower value of boiling point than H_2O
(C) D_2O has higher number of neutrons than H_2O
(D) H_2O has smaller number of protons than D_2O

16. Choose incorrect order regarding mentioned properties :

- (A) $\text{SeO}_2 > \text{SO}_2$ (Reducing power)
- (B) $\text{Na} > \text{K}$ (Softness of metal)
- (C) $\text{PbCl}_4 > \text{PbI}_4$ (stability)
- (D) $\text{H}_2\text{S} > \text{H}_2\text{O}$ (Acidic character)

17. Which of the reaction does NOT give product as indicated ? (ignore balancing)

- (A) $\text{PbS(s)} + \text{O}_3\text{(g)} \rightarrow \text{PbSO}_4\text{(s)} + \text{O}_2\text{(g)}$
- (B) $\text{ZnS(s)} + \text{O}_2\text{(g)} \rightarrow \text{ZnO(s)} + \text{SO}_2\text{(g)}$
- (C) $\text{Pb}_2\text{O}_3 + \text{HNO}_3 \rightarrow \text{Pb(NO}_3)_2 + \text{PbO} + \text{H}_2\text{O}$
- (D) $\text{KClO}_3\text{(s)} \xrightarrow{\text{MnO}_2} \text{KCl(s)} + \text{O}_2\text{(g)}$

18. Among statements (a) -(d), the correct ones are :

- (a) Lime stone is decomposed to CaO and combined with SiO_2 during the extraction of iron from its oxides.
- (b) In the extraction of silver, silver is leached out as an anionic complex.
- (c) Nickel is purified by Mond's process.
- (d) Zr and Ti are purified by Van Arkel method.
- (A) (c) and (d) only
- (B) (a), (c) and (d) only
- (C) (b), (c) and (d) only
- (D) (a), (b), (c) and (d)

19. Which of the following statements are correct ?
 (a) Both LiCl and MgCl_2 are soluble in ethanol.
 (b) The oxides Li_2O and MgO combine with excess of oxygen to give superoxide.
 (c) LiF is less soluble in water than other alkali metal fluorides.

(d) Li_2O is more soluble in water than other alkali metal oxides.

Choose the most appropriate answer from the options given below:

- (A) (a) and (c) only
- (B) (a), (c) and (d) only
- (C) (b) and (c) only
- (D) (a) and (c) only

20. The initial rate of hydrolysis of methyl acetate (1M) by a weak acid (HA , 1M) is $1/100^{\text{th}}$ of that of a strong acid (HX , 1M), at 25°C . The K_a of HA is

- (A) 1×10^{-4}
- (B) 1×10^{-5}
- (C) 1×10^{-6}
- (D) 1×10^{-3}

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

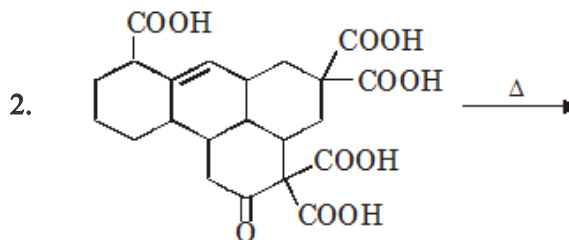
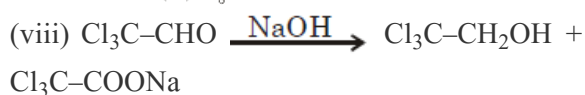
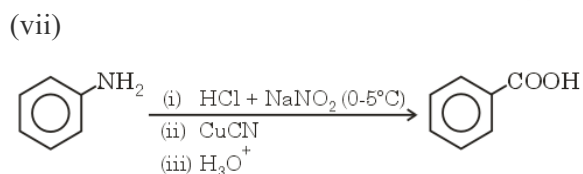
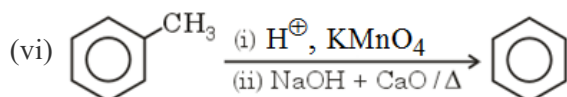
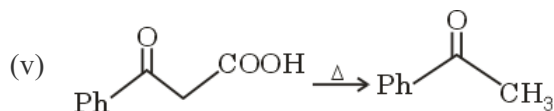
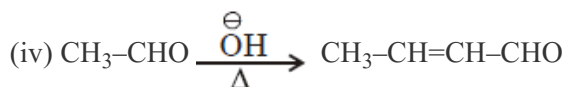
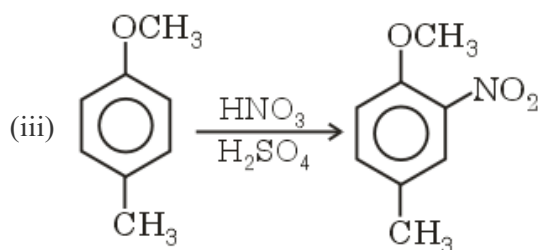
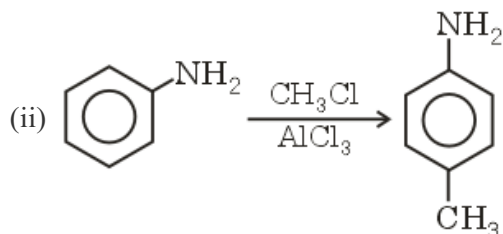
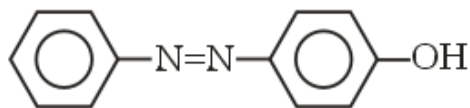
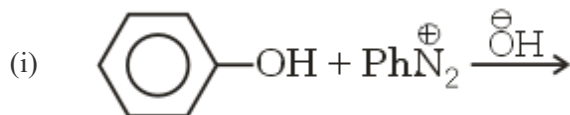
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. How many reaction correctly match with their major product :



Number of CO_2 evolved.

3. 50 mL of 0.1 M CH_3COOH is being titrated against 0.1 M NaOH. When 25 mL of NaOH has been added, the pH of the solution will be _____ $\times 10^{-2}$. (Nearest integer)

(Given : $\text{pK}_a (\text{CH}_3\text{COOH}) = 4.76$)

$\log 2 = 0.30$

$\log 3 = 0.48$

$\log 5 = 0.69$

$\log 7 = 0.84$

$\log 11 = 1.04$

4. Find $\Delta G(\text{J/mol})$ for the reaction at 300kPa & 27°C when all gases are in stoichiometric ratio of moles. $\text{N}_2\text{O}_4(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$ [$\ln 2 = 0.7$]

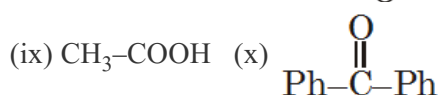
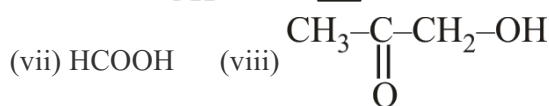
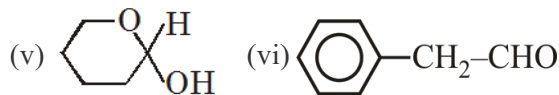
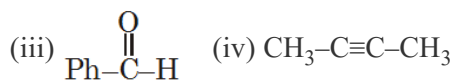
Given : $\Delta G_f^\circ (\text{N}_2\text{O}_4) = 100\text{kJ/mol}$

$\Delta G_f^\circ (\text{NO}_2) = 50\text{kJ/mol}$

$R = 8 \text{ J/mol-K}$

5. A solution containing $2.5 \times 10^{-3} \text{ kg}$ of a solute dissolved in $75 \times 10^{-3} \text{ kg}$ of water boils at 373.535 K . The molar mass of the solute is _____ g mol^{-1} . [nearest integer] (Given: $K_b (\text{H}_2\text{O}) = 0.52 \text{ K Kg mol}^{-1}$, boiling point of water = 373.15K)

6. Identify total number of compounds which give both tollen's and fehling test ?



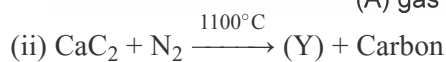
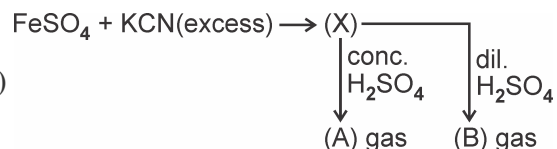
7. Match given properties with elements

	List-I (Properties)		List-II (Elements)
P	Produce stable M ₂ O type oxide with limited supply of air	1	Li, Be, Al
Q	Chloride of elements produce red colour in oxidising flame	2	Li, Ca, Sr
R	Chlorides of such elements having significant covalent character or covalent nature in anhydrous form	3	Na, Li, Cs

Match list-I and list-II and give your answer as abc where a,b and c are the match of 1, 2, 3 from list-II.

(for example Matching of P is = 2, Q = 3 & R = 1 then your answer is 231)

- 8.



Find the sum of the number of π bonds present in (A), (B) and (Y).

9. Match the colour with given compounds

	List-I (Compounds)		List-II (Colour)
P	Mn ₂ O ₇	1	Orange
Q	(NH ₄) ₂ Cr ₂ O ₇	2	Green
R	CrO ₂ Cl ₂	3	Reddish Brown

Match list-I and list-II and give your answer as abc where a,b and c are the match of 1, 2, 3 from list-II.

(for example Matching of P is = 2, Q = 3 & R = 1 then your answer is 231)

- 10.

	List-I Catalyst		List-II Process
(P)	V ₂ O ₅	(1)	Ostwald process
(Q)	CuCl ₂	(2)	Deacon process
(R)	Pt/Rh	(3)	Contact process

Match list-I and list-II and give your answer as abc where a,b and c are the match of 1, 2, 3 from list-II.

(for example Matching of P is = 2, Q = 3 & R = 1 then your answer is 231)

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. If the function f, g, h are defined from the sets of real numbers R to R such that $f(x) = x^2 - 1$, $g(x) = \sqrt{x^2 + 1}$, $h(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ x, & \text{if } x > 0 \end{cases}$, then the composite function $h \circ f \circ g(x)$ is

(A) $\begin{cases} 0, & x = 0 \\ x^2, & x > 0 \\ -x^2, & x < 0 \end{cases}$ (B) $\begin{cases} 0, & x = 0 \\ x^2, & x \neq 0 \end{cases}$
 (C) $\begin{cases} 0, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ (D) $\begin{cases} x^2, & x \leq 0 \\ -x^2, & x > 0 \end{cases}$

2. If $\vec{a}, \vec{b}$ are non-zero vectors such that $|\vec{a} + \vec{b}| = |\vec{a} - 2\vec{b}|$, then -

(A) $\vec{a} \cdot \vec{b} = 2|\vec{b}|^2$
 (B) $\vec{a} \cdot \vec{b} = |\vec{b}|^2$
 (C) Least value of $\vec{a} \cdot \vec{b} + \frac{4}{|\vec{b}|^2 + 2}$ is $2\sqrt{2}$
 (D) Least value of $\vec{a} \cdot \vec{b} + \frac{4}{|\vec{b}|^2 + 2}$ is $2\sqrt{2} - 1$

3. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$; $6A^{-1} = A^2 + cA + dI$ then (c, d) is

(A) $(-6, 11)$ (B) $(-11, 6)$
 (C) $(11, 6)$ (D) $(6, 11)$

4. Unit vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar, unit vector $\vec{d}$ is perpendicular to them. If $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \frac{i - 2j + 2k}{6}$ and angle between $\vec{a}$ and $\vec{b}$ is 30° then $\vec{c}$ can be

(A) $\frac{i - j + k}{\sqrt{3}}$ (B) $\frac{i - 2j + 2k}{3}$
 (C) $\frac{2i - j + k}{\sqrt{6}}$ (D) $\frac{i - 2j + 2k}{6}$

5. Consider the three linear equations with respect to X, Y and Z as

$x. X + Y + 2Z = 0$
 $f(x). X + 3Y + x^2Z = 0$
 $5x X + 6y + Z = 0$ having non-trivial solution for X, Y and Z then curve $f(x)$

(A) is always increasing
 (B) is always decreasing
 (C) has exactly one critical point
 (D) has three critical points

6. The value of definite integral $\int_{-\pi}^{\pi} (\cos 2x \cdot \cos 2^2x \cdot \cos 2^3x \cdot \cos 2^4x \cdot \cos 2^5x) dx$ is

(A) 1 (B) -1 (C) 2 (D) 0

7. If $\theta_1 = \sin^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{1}{3}\right)$ and $\theta_2 = \cos^{-1}\frac{4}{5} + \cos^{-1}\frac{1}{3}$ then

- (A) $\theta_1 < \theta_2$ (B) $\theta_1 > \theta_2$
(C) $\theta_1 = \theta_2$ (D) $\theta_1 - 2\theta_2$

8. The function $f: [a, \infty) \rightarrow \mathbb{R}$, where $\mathbb{R}$ denotes the range of $f(x)$ corresponding to the given domain $[a, \infty)$. If $f(x) = 2x^3 - 3x^2 + 6$ will have an inverse, then complete set of possible values of a is

- (A) $(-\infty, 1]$ (B) $[0, \infty)$ (C) $(-\infty, 0]$ (D) $[1, \infty)$

9. If $f(x) = \int e^x \left(\tan^{-1}(x) + \frac{2x}{(1+x^2)^2} \right) dx$, $f(0) = -1$ then the value of $f(1)$ is

- (A) $e\left(\frac{\pi}{4} - \frac{1}{2}\right)$ (B) $e\left(\frac{\pi}{4} + \frac{1}{2}\right)$
(C) $e\left(\frac{\pi}{2} - \frac{1}{4}\right)$ (D) $e^{-1}\left(\frac{\pi}{4} - \frac{1}{2}\right)$

10. If $\sqrt{x+y} + \sqrt{y-x} = c$ then $\frac{d^2y}{dx^2}$ equals

- (A) $\frac{2}{c^2}$ (B) $\frac{-2}{c^2}$ (C) $\frac{2}{c}$ (D) $\frac{-2}{c}$

11. If $I_n = \int (\operatorname{cosec} 2x)^n dx$ then

- (A) $I_n = \frac{\cot 2x \operatorname{cosec}^{n-2} 2x}{-2(n-1)} + \frac{n-2}{n-1} I_{n-2}$
(B) $I_n = \frac{\cot 2x \operatorname{cosec}^{n-2} 2x}{-2(n-1)} + \frac{n+2}{n-1} I_{n-2}$
(C) $I_n = \frac{\cot 2x \operatorname{cosec}^{n-2} 2x}{-(n-1)} + \frac{n-2}{n-1} I_{n-2}$
(D) $I_n = \frac{\cot 2x \operatorname{cosec}^{n-2} 2x}{-(n-1)} + \frac{n+2}{n-1} I_{n-2}$

12. If water is poured into an inverted hollow right circular cone whose semi-vertical angle is 30° , its depth increase at the rate of 1 cm/s. The rate at which the volume of water increase when the depth is 24 cm is equal to :

- (A) $150\pi \text{ cm}^3/\text{s}$ (B) $180\pi \text{ cm}^3/\text{s}$
(C) $192\pi \text{ cm}^3/\text{s}$ (D) $200\pi \text{ cm}^3/\text{s}$

13. The minimum distance between the curves $y = \tan x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $\left(x - 2 - \frac{\pi}{4}\right)^2 + y^2 = 1$ is

- (A) $\sqrt{5} + 1$ (B) $\sqrt{5} - 1$
(C) $\sqrt{2} - 1$ (D) 2

14. If $f(x) = 1 - x - x^3$, then the set of all real values of x satisfying the inequality

- $1 - f(x) - f^3(x) > f(1 - 5x)$ is
(A) $(-2, 0) \cup (2, \infty)$ (B) $(-\infty, -2)$
(C) $(1, 2)$ (D) $(0, 2)$

15. Area bounded by the curves $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$ and $y = 3x - 15$ is

- (A) $\frac{73}{2}$ (B) $\frac{73}{3}$ (C) $\frac{73}{6}$ (D) 73

16. If $\sum_{r=0}^n \frac{(-1)^r \cdot {}^nC_r}{(r+1)(r+2)(r+3)} = \frac{1}{a(n+b)}$, then $a + b$ is equal to

- (A) 2 (B) 4
(C) 5 (D) 6

17. If $\frac{2}{1!9!} + \frac{2}{3!7!} + \frac{1}{5!5!} = \frac{2^m}{n!}$, then which of the following is correct ? (Where $m, n \in \mathbb{N}$)

- (A) $n - m = 1$ (B) $n + m = 18$
(C) $m - n = 1$ (D) $m + n = 20$

18. $\lim_{x \rightarrow 0^+} \left((\sin x)^{\frac{1}{x}} + (\cos x)^{\frac{1}{x}} + (\operatorname{cosec} x)^x \right)$ is equal to
 (A) 2 (B) 4
 (C) 6 (D) 8
19. If $y(x)$ is the solution of the differential equation $\frac{dy}{dx} = -2x(y - 1)$ with $y(0) = 1$, then $\lim_{x \rightarrow \infty} y(x)$ equals
 (A) 2 (B) 1
 (C) 4 (D) 5
20. Let $\vec{v}$ be a vector which satisfies the condition $2\vec{v} + \vec{v} \times (\hat{i} + 2\hat{j}) = 2\hat{i} + \hat{k}$. If $3|\vec{v}| = \sqrt{m}$, then find the value of m ?
 (A) 8 (B) 10 (C) 6 (D) 7

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Let m be the minimum value of the function $f(x) = 2x^3 - 3x^2 - 12x + 6$ on the set $A = \{x : \ln x > \ln(x^2 - 3x + 3)\}$, then find $|m|$

2. Let $f : \mathbb{R} \rightarrow \left(0, \frac{\pi}{2}\right]$ defined by $f(x) = \operatorname{cosec}^{-1}(ax^2 - 2px + 4)$ where $a, p \in \mathbb{N}$ is surjective then smallest value of $2(a + p)$ is
3. Let A is a non-zero matrix of order '2' such that $A + \operatorname{adj} A = A^{-1}$ then determinant of $(2A^{-1})$ is equal to
4. Let $[a_{ij}]$ is a square matrix of order 2 where $a_{ij} \in \{0, 1, 2, 3, 4, 6\}$ then number of matrices A such that A^{-1} exists is λ then $\lambda - 300$ is (given that all elements of matrix A are distinct)
5. Let $\vec{a} = 2\hat{i} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = 4\hat{i} + 3\hat{j} + 7\hat{k}$. If $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{b} = \vec{c} \cdot \vec{b}$, then the value of $\vec{a} \cdot \vec{r} =$
6. If $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$ and $\vec{a} \wedge \vec{b} = \vec{c} \wedge \vec{b} = \vec{a} \wedge \vec{c} = \theta$ where $\vec{p} \wedge \vec{q}$ represents the angle between vectors $\vec{p}$ and $\vec{q}$ then maximum value of θ is $\frac{\pi}{k}$. The value of $4k^2$ is
7. $I = \int_0^2 (3x^2 - 3x + 1) \cos(x^3 - 3x^2 + 4x - 2) dx = p(\sin q)$ where $p, q \in \mathbb{N}$, then find $(p^2 + q^3)$.
8. $\sum_{i=0}^{100} {}^{150}C_i \cdot {}^{350}C_{100-i} (150 - i) = k {}^{500}C_{100}$ then value of k is equal to
9. The last digit of $3^{81^n} + 1$, $n \in \mathbb{N}$, is
10. If $\vec{a}$ and $\vec{b}$ are non collinear vectors such that $(2x - y)\vec{a} + 4\vec{b} = 5\vec{a} + (x - 2y)\vec{b}$ then $(x + y)$ is

Space for Rough Work

